

Frequency transfer via a two-way optical phase comparison on a multiplexed fiber network

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We performed a two-way remote optical phase comparison on optical fiber. Two optical frequency signals were launched in opposite directions in an optical fiber and their phases were simultaneously measured at the other end. In this technique, the fiber noise was passively cancelled, and we compared two optical frequencies at the ultimate 10^{-21} stability level. The experiment was performed on a 47 km fiber that is part of the metropolitan network for Internet traffic. The technique relies on the synchronous measurement of the optical phases at the two ends of the link, that is made possible by the use of digital electronics. This scheme offers several advantages with respect to active noise cancellation, and can be upgraded to perform more complex tasks.

The advent of optical clocks has enabled frequency metrology to achieve the 10^{-18} level of accuracy [1]. These extremely high performances pave the way for a number of applications in fundamental physics, high resolution spectroscopy [2] and relativistic geodesy [3], but at the same time, they require an adequate technique to perform frequency comparisons between distant clocks. Phase-compensated optical links have proved to be reliable from this point of view and outperform state-of-the-art satellite techniques by orders of magnitude [4]. The transmission of RF and microwaves [5–7], optical frequencies [8–10], and of an optical comb [11] have been demonstrated, and time dissemination has recently been performed as well [12–14]. In coherent optical links the phase noise added by the fiber due to environmental noise is actively cancelled. This is obtained by delivering an optical signal to the remote end and by reflecting a part of the transmitted radiation back to the local laboratory. Here the round trip signal is compared to the original one, and the phase noise added by a double pass in the fiber is detected and compensated with a phase locked loop (PLL). Active noise cancellation allows the delivery of an optical frequency over hundreds of kilometers, with a resolution at the 10^{-20} level. However, the bidirectional operation of the link poses several issues: the PLL bandwidth is limited by the round trip time of light in the fiber. In addition, even if the locking bandwidth is large, the PLL stabilizes the phase of the round-trip signal, whilst the stabilization of the one-way signal is desirable [15]. Besides this fundamental limitation, other technical issues exist: in real implementations, the beatnote between the local and the round-trip signal is often deteriorated by undesired backreflections, optical and electrical wideband noise, and is detected with a poor signal to noise ratio (SNR). In addition, amplitude modulation may occur, especially if optical amplifiers are used along the way. In most cases, a clean-up tracking oscillator is required, to filter the wideband noise and eliminate amplitude modulation. If the SNR at detection is low, however, the clean-up oscillator is affected by

cycle slips, which result in glitches and possible frequency biases on the delivered signal [16]. Cycle slips may also happen with non-stationary noise events, or if the tracking oscillator bandwidth is too low.

In this work, we investigate an alternative technique for comparing distant ultrastable lasers that does not require the active fiber noise cancellation. The noise is cancelled by data post-processing, in analogy to two-way methods, such as satellite links for frequency transfer [4]. This technique relies on the synchronous phase-comparison of two ultrastable lasers at the two fiber ends. The lasers must have a coherence length which is longer than the fiber haul. They are injected in the link at the two opposite ends and travel along the fiber, and their optical phases are measured at the other end against the local laser. If the same fiber is used in both directions, and the phase measurement is synchronous at the two ends, the link noise is cancelled when comparing the two datasets. The main requirement is the synchronous phase-comparison: in our system, two Tracking Direct Digital Synthesizers (Tracking DDSs) [17] measure the optical phases at each fiber end with negligible delay and no dead time [18, 19].

Digital implementation is reliable and can be upgraded to perform other tasks such as time dissemination, with reduced costs and easier replication than modem-based systems. Furthermore, this setup may be useful for novel applications of fiber links, such as the investigation of non reciprocal effects in large fiber loops [20].

This Letter describes the optical and the electronic systems. Then, it reports on the results, highlighting advantages and limitations of this technique.

In this work we compared two ultrastable lasers at 194 THz separated by a 47 km fiber, that is part of the metropolitan fiber network. This fiber is used for the Internet data traffic and is implemented on a Dense Wavelength Division Multiplexed (DWDM) architecture, with ~ 23 dB of optical losses. The 44th channel of the International Telecommunication Union (ITU) grid (wavelength 1542.14 nm) has been dedicated to our experi-

ment, while Internet data are transmitted on the 21st and 22nd channel, 2 THz away. The fiber has both ends in our laboratory, and the same laser was used in the two directions, to investigate the ultimate resolution of this scheme. The setup is sketched in Figure 1. An ultrastable frequency signal at 194 THz was provided by a fiber laser frequency locked with the Pound-Drever-Hall technique to a Fabry-Perot high-finesse cavity (120,000) made of Corning Ultra Low Expansion (ULE) glass. The resulting laser linewidth is less than 30 Hz [21]. The laser was split into two parts that simulated two different lasers located in distant laboratories. At each side, part of the radiation was injected into the fiber, while the remaining radiation served as a local oscillator. We used two Acousto-Optic Modulators (AOMs) at about 40 MHz frequency-separated by nearly 500 kHz, to distinguish the signal coming from the far fiber end from the stray reflections. Two Optical Add&Drop Multiplexers were used to inject and extract our signal from the multiplexed fiber network. At each side, the beatnote between the local and the received light was detected with a photodiode, filtered and amplified; then, its phase φ_a (φ_b) was measured with a tracking system based on a DDS. The scheme of the Tracking DDS is that of a classical PLL, where the Voltage Controlled Oscillator (VCO) is replaced by a DDS. The phase discriminator is a double balanced mixer. Its output is digitalized through an Analog to Digital Converter (ADC) and fed to a servo, that calculates the correction for the DDS. Within the PLL bandwidth, the sequence of data sent to the DDS coincides with the tracking phase. This data stream gives direct access to the beatnote phase, without the need for additional instrumentation such as Fast Fourier Transform Spectrum Analyzers, or phase/frequency meters. A Field Programmable Gate Array (FPGA) implements the PLL controllers and guarantees the synchronization at the μ s level. Moreover it unambiguously sets the measurement bandwidth and decimates data to fit the PC link bandwidth, also reducing the storage requirements. In addition, DDSs have a wide output frequency range, that provides additional flexibility to the experiment.

This system has a tracking bandwidth of about 20 kHz,

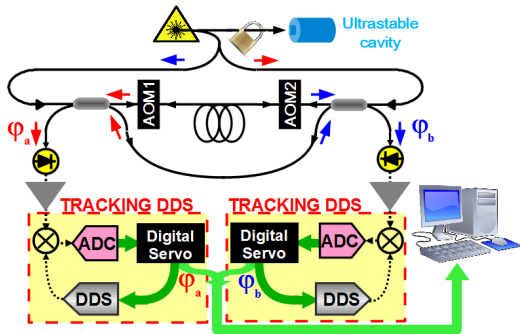


FIG. 1. The optical apparatus and the electronic system: AOM Acousto Optic Modulators, ADC Analog to Digital Converters, DDS Direct Digital Synthesizers.

limited by the serial driving of the DDS. This is the minimum bandwidth required by this link, as demonstrated by the presence of some cycle slips (about 10 per hour). By parallel driving the DDS, a bandwidth of up to 1 MHz is feasible, that is suitable for hauls of hundreds kilometers. In practice, the tracking bandwidth of the DDSs must be such that the phase error at closed loop is minimized, to prevent cycle-slips [16]. Hence, it must be adequate to track the fiber acoustic noise but, at the same time, the phase noise floor must be sufficiently low. Thus, the real limitation to the tracking bandwidth is the SNR at detection. In this sense, as is explained below, the two-way scheme is advantageous, allowing full benefit of the 1 MHz bandwidth. Figure 2 shows the time evolution of the beatnote phases as measured at the two fiber ends, and their difference. The fiber accumulated about 60 ps in 50000 s, and noise was cancelled at the 0.1 fs level when calculating the difference. The initial transient and the residual noise of the phase difference were due to slow temperature changes of the laboratory, that affected the short, non common optical fibers of the interferometer. The glitches appearing on the phase difference were due to occasional cycle slips. They have been removed off-line and were not an issue in the analysis.

Figure 3 shows the phase noise spectral density (PSD)

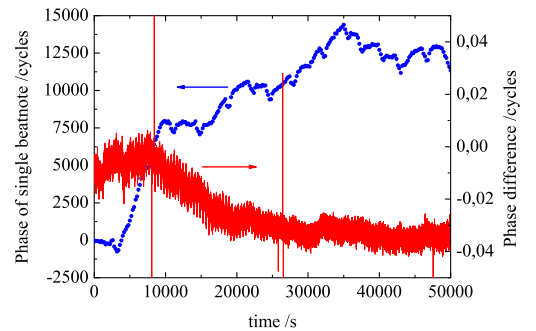


FIG. 2. The phase of one of the two independent beatnotes (left-hand axis, blue circles) and of their difference (right-hand axis, red line). 1 cycle is about 5 fs.

of one of the two beatnotes (blue line) and of the phase difference (red line). At Fourier frequencies $f < 1$ Hz the noise was due to the short fibers that were not common in the two systems. At $f > 1$ Hz, the noise was due to optical length variations that were not correlated over the entire fiber. The graph shows their expected contribution (black line), assuming that all the noise sources were spatially uncorrelated.

With this technique the noise compensation is not limited by the round trip time, as there is not an active noise cancellation loop. In principle, for noise sources that are correlated along the whole fiber length L (e.g. temperature), the ultimate performances could be better than in actively stabilized links. However, such noise sources

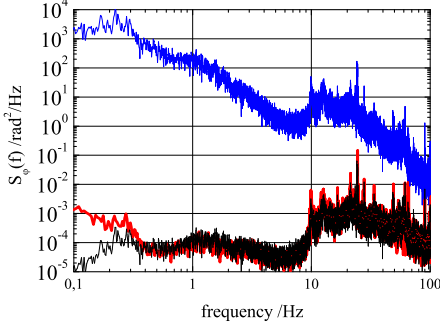


FIG. 3. The phase noise spectral density of the one-way signal (blue line), of the phase difference (red line) and the expected limitation due to uncorrelated noise (black line).

affect only minorly real links, where most of the noise is spatially uncorrelated. The contribution for spatially-uncorrelated noise, integrated over the whole fiber, can be evaluated considering the phase variation $\delta\varphi_F(z, t)$ in each fiber segment as a function of position z and time t . Since two counterpropagating beams travel along each position at different times, their phase difference $\delta\varphi_D$ at the output is:

$$\delta\varphi_D(t) = \int_0^L \delta\varphi_F(z, t - \tau + \frac{z}{c}) dz - \int_0^L \delta\varphi_F(z, t - \frac{z}{c}) dz$$

where $\tau = n \frac{L}{c}$ and n is the refractive index of the fiber. After integration, the PSD $S_{\varphi,D}(f)$ of the phase difference can be computed:

$$S_{\varphi,D}(f) = \frac{1}{3}(2\pi\tau f)^2 S_{\varphi,F}(f) \quad (1)$$

where $S_{\varphi,F}(f)$ is the PSD of the one-way fiber noise. Eq. 1 holds in the spectral region where $2\pi f\tau \ll 1$. The expected contribution under the assumption that all the noise was spatially uncorrelated, shown in Figure 3, is in good agreement with the experimental results.

Phase data can be differentiated to obtain the instantaneous beatnote frequency on both link ends. The mean frequency difference of the two beatnotes was $< 4 \times 10^{-21}$. The stability of the frequency difference is shown in Figure 4 in terms of Allan deviation $\sigma_y(t_a)$ as a function of the averaging time t_a . It achieves 4×10^{-21} at 10^4 s, on a measurement bandwidth of 0.5 Hz.

It is interesting to estimate the performance deterioration due to a not perfect synchronization of the samples. Following the same approach used to derive eq. 1, the time mismatch can be modelled as an additional delay δ ; τ is then replaced by $\tau + \delta$. After some algebraic manipulation, and again assuming that all the noise is uncorrelated along the fiber, one ends up with:

$$S_{\varphi,D}(f) = \left(\frac{1}{3}(2\pi\tau f)^2 + (2\pi\delta f)^2 \right) S_{\varphi,F}(f), \quad (2)$$

holding in the spectral region where $2\pi f(\tau + \delta) \ll 1$. A factor $3\left(\frac{\delta}{\tau}\right)^2$ deterioration is expected with respect to

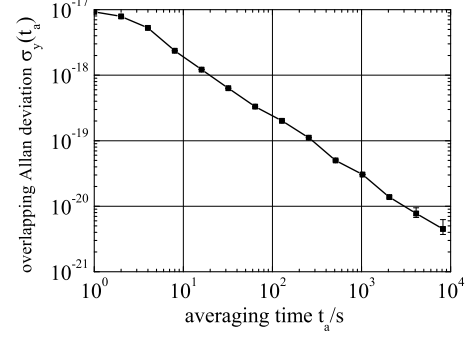


FIG. 4. The overlapping Allan deviation $\sigma_y(t_a)$ for the frequency difference on a bandwidth of 0.5 Hz.

the optimal case, if a delay $|\delta| > \tau$ is introduced.

Again, this model was confirmed by the experimental data. We evaluated the phase noise increase of the phase difference for several values of δ . Figure 5 shows the value of the quantity $\rho = \sqrt{S_{\varphi,D}(f)/S_{\varphi,D}^0(f)}$ at $f = 1$ Hz, where $S_{\varphi,D}(f)$ denotes the PSD of the phase difference with delayed samples, and $S_{\varphi,D}^0(f)$ denotes the PSD of the phase difference with synchronously subtracted samples. The graph shows the obtained points (black squares) and the calculated value (line), according to eq. 2, as a function of $|\delta/\tau|$.

In practice, timing at the μ s level is feasible, and does

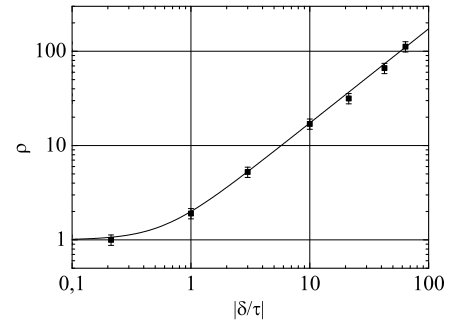


FIG. 5. Measured (squares) and calculated (line) degradation of the phase noise when an additional delay δ due to a bad synchronization is included. The graph shows $\rho = \sqrt{S_{\varphi,D}(f)/S_{\varphi,D}^0(f)}$ at $f = 1$ Hz, where $S_{\varphi,D}(f)$ and $S_{\varphi,D}^0(f)$ are the PSD of delayed and synchronously subtracted phase samples, as a function of $|\delta/\tau|$.

not require continuous monitoring, as typical delay variations are negligible at this level [12]. Synchronization at the μ s level is widely enough even for a short 47 km link (in which $\tau = 235 \mu$ s), with a noise increase below 1%. Timing becomes less stringent for longer links. If data postprocessing is used, an algorithm can also be developed to minimize $S_{\varphi,D}(f)$, avoiding the need for a precise synchronization. This may be helpful if some

segments of the link are much noisier than others, as the algorithm can be optimized to cancel their contribution. In summary, we implemented a two-way optical frequency transfer system on optical fiber, based on the remote synchronous measurement of the optical phase, and demonstrated its performance at the 10^{-21} level of stability. This scheme offers some advantages with respect to traditional active noise cancellation schemes, when a clock comparison and no frequency dissemination is needed. Since two beams travel simultaneously in the two directions, the difference of the two beatnotes is at first order insensitive to the fiber noise. Passive noise compensation is better from a conceptual point of view, as a better cancellation is expected for the noise that is correlated over the whole fiber length. Moreover, some technical limitations of actively compensated fiber links become less stringent, as in this setup each beam travels only once in the fiber. The advantages include: a half phase noise of the fiber; a lower wideband noise of optical amplifiers, and less amplitude modulation of the optical carrier; better optical power and SNR at

the two ends, that may allow to use less amplifiers and enables a higher tracking bandwidth. These aspects are especially desirable in long optical links, and enable a better signal tracking and significantly less cycle slips. The digital architecture allows fast tuning, and most system upgrades are feasible just with additional firmware. Moreover, the data of interest, such as phase, frequency, signal power, are routinely measured inside the FPGA, and can be monitored without any external instrumentation. Thus, this scheme is suitable for autonomous and remotely controlled link operation.

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